St. Philomena's College (Autonomous),
Bannimantap, Mysuru - 15.

PG Department of Mathematics

A Report On

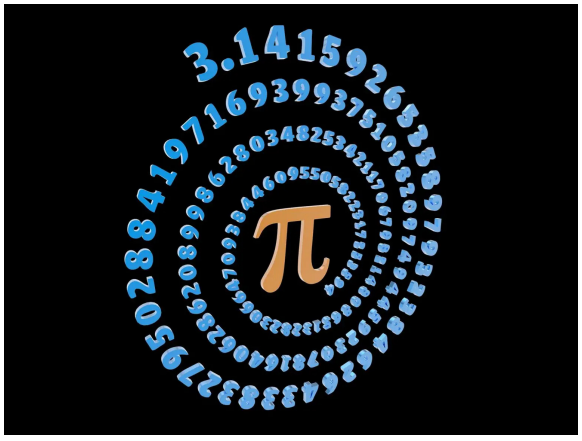
INTERACTIVE SESSION ON THE OCCASION OF PI DAY

held on 14-03-2023

The Essence of Mathematics

Resource Person: Mr. Arun Kumar A,
Special Officer,
Office of Regional Joint Director,
DCE, Mysuru.

Venue: Conference Hall, PG Block.



Objectives

1) History of Number System.

2) To give insights on formulation of π.

3) To explain real-life applications of real numbers.

4) To discus the challenges and pertinent open problems.

Outlines of the Session:

• Preliminaries on Number System.

• Origin and history of an irrational number π.

• Mentioning the significance of Archimedean Principle.

• Few applications of Real numbers in Astrophysics, Nonlinear Dynamics, Chemical Graph theory.

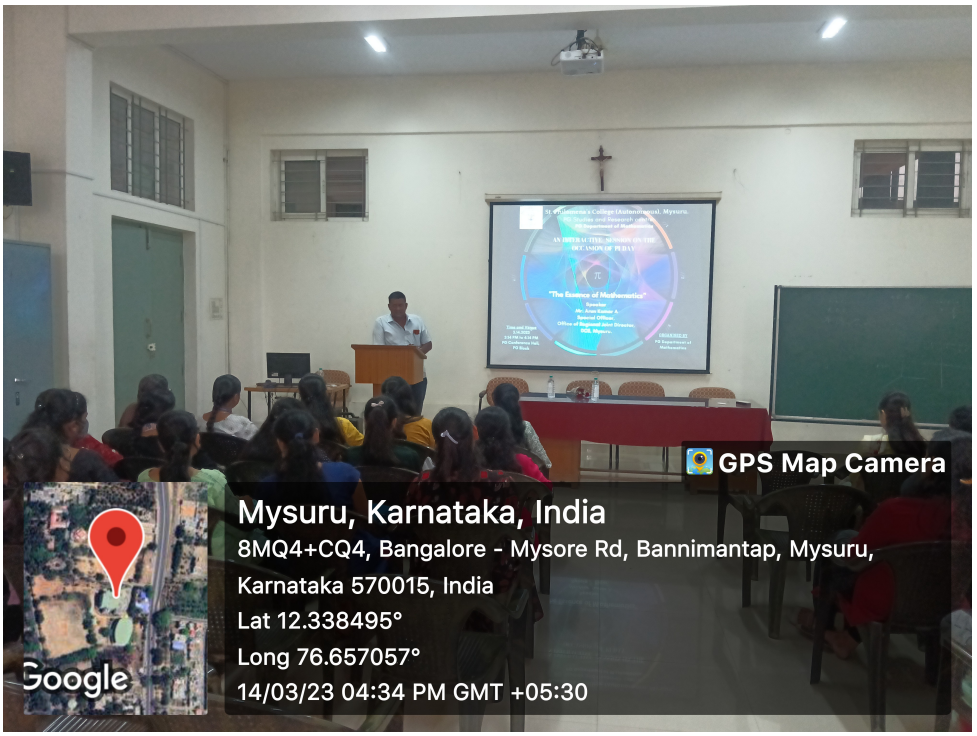
• Open problems like formulation of cubic formula to deduce the roots of polynomial of degree three, Twin Prime Conjecture, Goldbach's Conjecture, Diophantine Equations and NP hard problems.



The Mathematical interactive session on "The Essence of mathematics" by Arun Kumar A was held on the occasion of Pi Day, 14th of March 2023. The speaker Mr Arun Kumar A is a Special Officer, Office of Regional Joint Director, DCE, Mysuru. The speaker is a former Assistant Professor, GFGC Bannur, Mysuru and our proud alumni having vast experience in teaching field. Mr. Bharatha K, Assistant Professor, PG Department of Mathematics, welcomed the gathering and introduced the speaker under the presence of our beloved Father Christopher. The session began soon after welcoming the speaker.



Mr. Arun Kumar started the session by discussing the basic concepts of Number Systems and the necessity of classification of numbers. He described the history of π is our ancient scriptures like Rigvedha, where the value of π was correctly calculated and coded in form of Sanskrit Slokas. He briefly explained the significance of π in mathematics and allied fields over years since its origin and gave some insights about the origin of this irrational number π. The speaker then spoke about the essence and mathematical meaning of the Archimedean Principle and made students to interact with few thoughtful mathematical problems.



Moving on, he discussed about real-life applications of real numbers in different fields namely Astrophysics, Nonlinear Dynamics and Chemical Graph theory through some examples. He then dealt with few mathematical conjectures which are open till today like cubic formula to deduce the roots of polynomial of degree three, Twin Prime Conjecture, Goldbach's Conjecture, Diophantine Equations and NP hard problems and gave insights about how several mathematicians have looked up to these topiocs in the past years.

He concluded the session by motivating the students to study mathematics by understanding its essence and enjoying its beauty.